

The empirical recurrence for OEIS A260542, recovered from its tail

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Abstract

OEIS A260542 records “Empirical recurrence of order 91” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 148$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 96$. Its last coefficient is $c_{91} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-91 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260542 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000011 or 00010001.”. It has offset 1 and begins

1350, 2229, 5592, 10949, 21050, 50674, 105058, 215160, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:09:02, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 148$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 148$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 91: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 91.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 296$ exact terms from $n = 5$ on, it returns order 91 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	0	c_{24}	41177	c_{47}	23194	c_{70}	102634
c_2	1	c_{25}	102409	c_{48}	−573555	c_{71}	−17302
c_3	12	c_{26}	85160	c_{49}	425500	c_{72}	−49257
c_4	19	c_{27}	−8770	c_{50}	170527	c_{73}	60293
c_5	−11	c_{28}	−76523	c_{51}	−135674	c_{74}	−32285
c_6	−69	c_{29}	−187936	c_{52}	581468	c_{75}	5270
c_7	−171	c_{30}	−125353	c_{53}	−470789	c_{76}	13013
c_8	−119	c_{31}	14938	c_{54}	146458	c_{77}	−15638
c_9	261	c_{32}	95392	c_{55}	94478	c_{78}	7604
c_{10}	807	c_{33}	276049	c_{56}	−523928	c_{79}	−1027
c_{11}	1068	c_{34}	163204	c_{57}	582774	c_{80}	−2216
c_{12}	216	c_{35}	31276	c_{58}	−386981	c_{81}	2524
c_{13}	−2362	c_{36}	−59785	c_{59}	3711	c_{82}	−1172
c_{14}	−4882	c_{37}	−315912	c_{60}	403692	c_{83}	176
c_{15}	−3822	c_{38}	−185176	c_{61}	−541643	c_{84}	220
c_{16}	1573	c_{39}	−105881	c_{62}	380973	c_{85}	−253
c_{17}	12233	c_{40}	−135118	c_{63}	−49209	c_{86}	102
c_{18}	18267	c_{41}	339902	c_{64}	−252688	c_{87}	−9
c_{19}	8036	c_{42}	242249	c_{65}	353520	c_{88}	−8
c_{20}	−12559	c_{43}	109695	c_{66}	−236466	c_{89}	9
c_{21}	−42203	c_{44}	419477	c_{67}	40758	c_{90}	−5
c_{22}	−46073	c_{45}	−415425	c_{68}	127551	c_{91}	1
c_{23}	−6343	c_{46}	−301735	c_{69}	−169897		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 96$, and it is the recurrence OEIS A260542 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{91} - c_1x^{91-1} - \dots - c_{91}$. Its constant term is $-c_{91} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 91 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 91, so $\deg q = 91$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 91, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 91, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-91 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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